

Data Analytics and Mining

Intro and Course overview

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September 18, 2026

Teaching Team Contacts / Timetable

- ▶ Fedor Shlyakhov (lectures and seminars)
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Lectures	Seminars	Format	Duration
Fridays 18:10	Fridays ~ 19:40	Offline	Modules 1-2

Repository with course material

<https://github.com/FedorShlyakhov/DAM2026>

Course content

- ▶ Introduction to Data Analytics and Mining
 - Overview of data analytics and mining
 - Importance and applications
 - Types of data, structured vs. unstructured
- ▶ Data Processing and Cleaning
 - Handling missing data, outliers, noise
 - Data transformation, normalization, standardization, encoding
 - Feature selection and dimensionality reduction
- ▶ Exploratory Data Analysis (EDA)
 - Descriptive Statistics and Data Summarization
 - Data Visualization Techniques (Histograms, Scatter Plots, Box Plots)
 - Identifying Patterns, Trends, and Anomalies

Course content

- ▶ Statistical Foundations for Data Analysis
 - Probability Theory and Distributions
 - Correlation and Causation, introduction to Regression Analysis
- ▶ Machine Learning Fundamentals
 - Supervised vs. Unsupervised Learning
 - Linear and Logistic Regression
 - Support Vector Machines
 - Model Evaluation (accuracy, precision, recall, F1-score), Cross-Validation
- ▶ Data Mining Techniques
 - Classification, decision tree, k-Nearest Neighbors
 - Clustering, k-Means

Course content

- ▶ Advanced-Data Mining and Machine Learning
 - Ensemble Methods: Random Forests

Overview

- ▶ Compulsory course for year 2
- ▶ Duration: 1st half of the academic year (modules 1 and 2)
- ▶ Assessment elements:
 - Quizzes (50% weight)
 - Exam (50% weight), in the form of a project, with **progress tracked during the semester** (topic choice deadline, preliminary results deadline, final result deadline)
- ▶ Format:
 - Offline (lectures & seminars)

Grade Formula

Grade Component	Percentage	Evaluation Criteria
Quizzes	50%	In-class mini tests once a week or once in two weeks, each test consists of several questions, and a single test is a 10-point scale. IMPORTANT: Quizzes cannot be retaken. The only exception is if you are unable to attend due to illness and provide a medical certificate verified by the study office.
Final Project	50%	The final project is a 10-point scale (in groups), choose a dataset, make relevant analysis, write a report, and present their works.

The formula

$$\text{Final grade} = 0.5 \cdot \text{Quizzes} + 0.5 \cdot \text{Final Project}$$

$$0 \leq \text{Quizzes} \leq 10$$

$$0 \leq \text{Final Project score} \leq 10$$

Rounding to the closest integer

Arithmetic rounding. E.g. 3.5 is rounded to 4, 3.49 is rounded to 3.

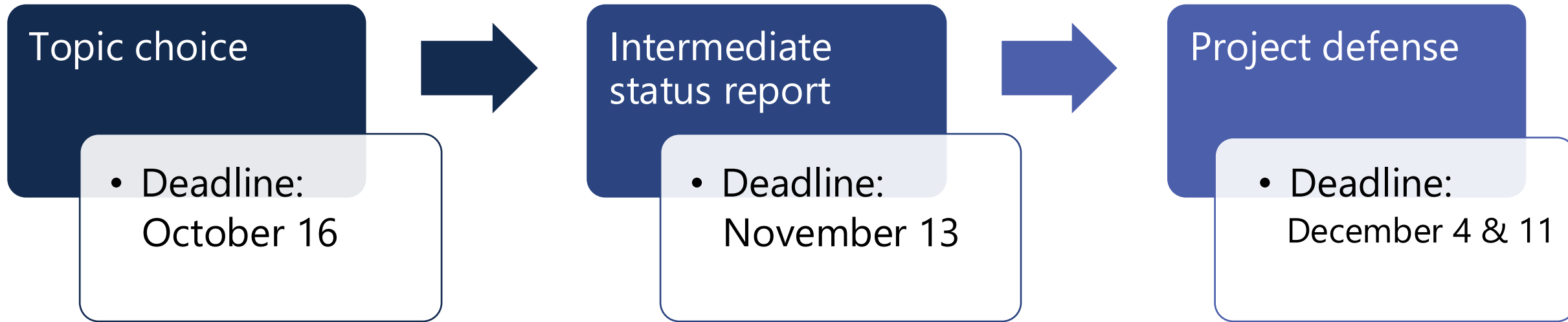
Quiz

- ▶ Quizzes are conducted during the seminars
 - ▶ **Format:** Quizzes will be taken in the LMS using the Safe Exam Browser (SEB)
 - ▶ **Devices:** Please use only your main device for the quiz. Other devices like phone, tablet, including AI tools, are not allowed.
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- **Note:** To keep things fair for everyone, any violation of these rules will unfortunately result in a score of 0 for that quiz.

Exam

- ▶ Exam in the form of project defense
- ▶ The project is:
 - Participation in a competition (on www.kaggle.com or similar)
 - Teams of up to 3 people are OK (roles of all members of a team should be clear and significant)
- ▶ **Please discuss your choice with me**

Exam project timeline



- ▶ Missing deadlines for stage 1, 2 will result in -1 point for the project (if both stage 1 and 2 were missed then -2 applies)
- ▶ Missing the project defense (stage 3) will result in 0 for the whole project.
- ▶ All members of a group need to attend the final defense.

Motivation

- ▶ Statistics and data analysis underlie machine learning and artificial intelligence (AI) technologies
- ▶ Comprehension of its basic principles helps to understand the world we live in

https://www.youtube.com/watch?v=RNnZwvklwa8&ab_channel=AdamEubanks